causal forest

2026-09-30

Setup

```
required_packages <- c("haven", "estimatr", "dplyr", "modelsummary")
missing_packages  <- setdiff(required_packages, rownames(installed.packages()))
if (length(missing_packages) > 0) {
  install.packages(missing_packages, repos = "https://cloud.r-project.org")
}
library(haven)
library(estimatr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(modelsummary)
library(tidyr)
library(grf)
library(knitr)


data_path <- file.path("data", "oreopoulos-resume-study-replication-data-file.dta")
if (!file.exists(data_path)) {
  stop("Data file not found. Expected it at: ", normalizePath(data_path, mustWork = FALSE))
}
df <- read_dta(data_path)
```

```
vars <- c("type", "female", "ba_quality", "extracurricular_skills", "language_skills",
          "ma", "same_exp", "exp_highquality", "reference", "accreditation", "legal")

df_int1 <- df %>%
  filter(
    type %in% c(0, 1),
    name_ethnicity %in% c("Canada", "Chinese", "Indian", "Pakistani")
  )
```

Check for missing values across waves (fall_data)

```
counts_all <- df_int1 %>%
  mutate(across(all_of(vars), as.character)) %>%
  pivot_longer(all_of(vars), names_to = "variable", values_to = "value") %>%
  mutate(variable = factor(variable, levels = vars)) %>%
  group_by(fall_data, variable) %>%
  summarise(
    with_value = sum(!is.na(value) & trimws(value) != ""),
    empty      = sum(is.na(value) | trimws(value) == ""),
    .groups = "drop"
  ) %>%
  left_join(df_int1 %>% count(fall_data, name = "total_rows"), by = "fall_data") %>%
  relocate(total_rows, .after = fall_data) %>%
  arrange(fall_data, variable)

print(counts_all, n = Inf)
```

```
## # A tibble: 33 x 5
##    fall_data total_rows variable               with_value empty
##        <dbl>      <int> <fct>                       <int> <int>
##  1         0       1902 type                         1902     0
##  2         0       1902 female                       1902     0
##  3         0       1902 ba_quality                   1902     0
##  4         0       1902 extracurricular_skills       1902     0
##  5         0       1902 language_skills              1902     0
##  6         0       1902 ma                           1902     0
##  7         0       1902 same_exp                     1902     0
##  8         0       1902 exp_highquality              1902     0
##  9         0       1902 reference                       0  1902
## 10         0       1902 accreditation                   0  1902
## 11         0       1902 legal                           0  1902
## 12         1       1495 type                         1495     0
## 13         1       1495 female                       1495     0
## 14         1       1495 ba_quality                   1495     0
## 15         1       1495 extracurricular_skills       1495     0
## 16         1       1495 language_skills              1495     0
## 17         1       1495 ma                           1495     0
## 18         1       1495 same_exp                     1495     0
## 19         1       1495 exp_highquality              1495     0
## 20         1       1495 reference                       0  1495
## 21         1       1495 accreditation                   0  1495
## 22         1       1495 legal                           0  1495
## 23         2       2260 type                         2260     0
## 24         2       2260 female                       2260     0
## 25         2       2260 ba_quality                   2260     0
## 26         2       2260 extracurricular_skills       2260     0
## 27         2       2260 language_skills              2260     0
## 28         2       2260 ma                           2260     0
## 29         2       2260 same_exp                        0  2260
## 30         2       2260 exp_highquality              2260     0
## 31         2       2260 reference                    2260     0
```

```
## 32         2       2260 accreditation                2260     0
## 33         2       2260 legal                        2260     0
```

```
df_int1 <- df_int1 %>%
  mutate(
    D              = as.numeric(type == 1),   # 1 = foreign name (Type 1), 0 = Type 0 baseline
    fall_data      = factor(fall_data),
    female                 = ifelse(is.na(female), 0, female),
    ba_quality              = ifelse(is.na(ba_quality), 0, ba_quality),
    extracurricular_skills  = ifelse(is.na(extracurricular_skills), 0, extracurricular_skills),
    language_skills         = ifelse(is.na(language_skills), 0, language_skills),
    ma                      = ifelse(is.na(ma), 0, ma),
    same_exp                = ifelse(is.na(same_exp), 0, same_exp),
    exp_highquality         = ifelse(is.na(exp_highquality), 0, exp_highquality)
    # Reference, accreditation, and legal variables are only available in some waves
    # (present only when fall_data == 2), so drop them to ensure consistency across waves.
  )

cat("Analysis sample size:", nrow(df_int1), "\n")
```

```
## Analysis sample size: 5657
```

```
cat("Treatment (D) distribution:\n")
```

```
## Treatment (D) distribution:
```

```
print(table(df_int1$D))
```

```
##
##    0    1
## 3026 2631
```

Causal forest 1 (3 covariates)

1. Cluster by firmid
2. X (Covariates): type, language, extracurricular, multinational
3. Y: callback rate
4.

```
forest1df_X <- df_int1 %>%
  select(language_skills,extracurricular_skills,same_exp)%>%
  as.matrix()

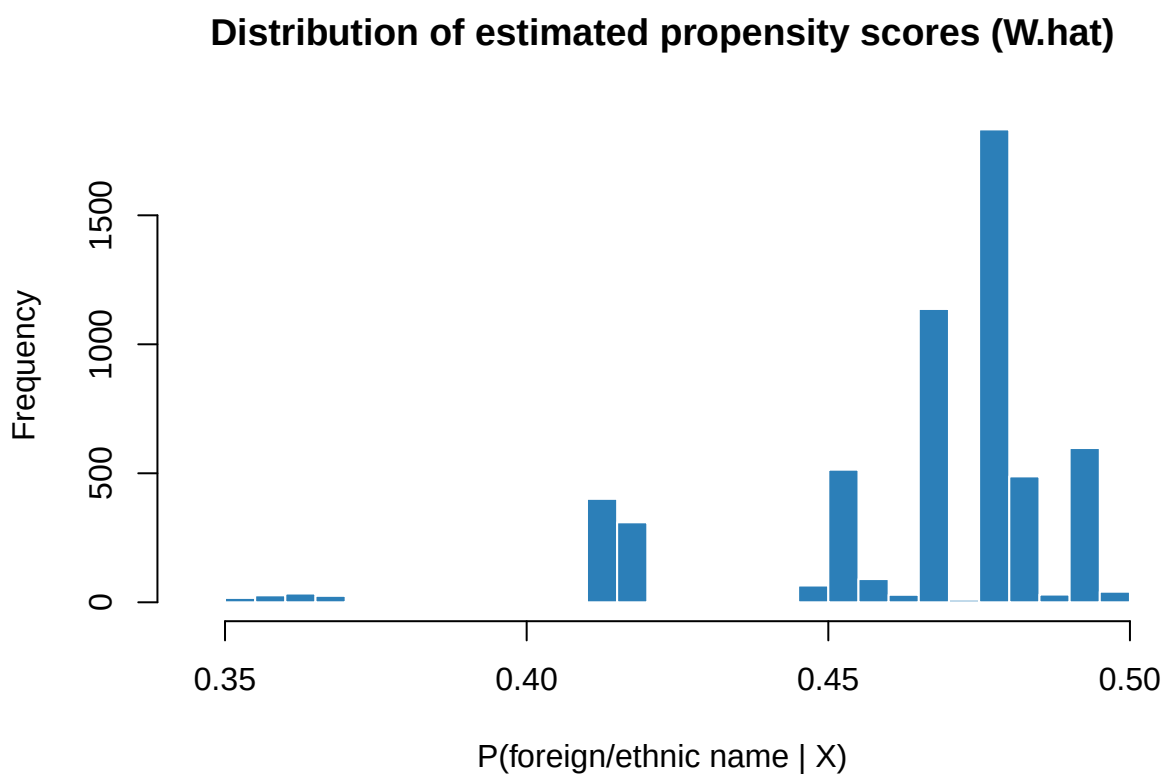
forest1df_treatment <- df_int1$D %>% as.matrix()
forest1df_cluster<-df_int1$firmid %>% as.matrix()
forest1df_Y<-df_int1$callback %>% as.matrix()

cf1<-causal_forest(forest1df_X,
                   forest1df_Y,
                   forest1df_treatment,
                   clusters=forest1df_cluster)
```

Overlap / propensity check

Because name ethnicity is experimentally randomized within a job posting, the estimated propensity scores should cluster fairly tightly around the actual treatment share rather than displaying the extreme values that signal confounding.

```
hist(cf1$W.hat,
     breaks = 30,
     col = "#2c7fb8",
     border = "white",
     main = "Distribution of estimated propensity scores (W.hat)",
     xlab = "P(foreign/ethnic name | X)")
```



Average treatment effect

```
ate <- average_treatment_effect(cf1, target.sample = "all")
ate
```

```
##     estimate      std.err
## -0.040968748  0.006880147
```

```r
cat(sprintf(
  "**Estimated ATE:** having a foreign/ethnic-sounding name changes the callback probability by **%.3f**
  100 * ate["estimate"], 100 * ate["std.err"]
))
```

Estimated ATE: having a foreign/ethnic-sounding name changes the callback probability by **-4.097** percentage points (SE = 0.688), on average across the analysis sample.

Variable importance

Which of the three moderators does the forest use most to split on (i.e., which one drives the most predictable heterogeneity)?

```r
vi <- variable_importance(cf1)
vi_tab <- data.frame(
  variable = colnames(forest1df_X),
  importance = as.numeric(vi)
) %>% arrange(desc(importance))

kable(vi_tab, digits = 3, caption = "Causal forest variable importance")
```

Table 1: Causal forest variable importance

variable	importance
extracurricular_skills	0.370
same_exp	0.317
language_skills	0.268

Causal forest 2 (ALL covariates)

1. Cluster by firmid
2. X (Covariates): ALL
3. Y: callback rate
4.

```r
# TO ASK: MUST WE INCLUDE FALL_DATA?
df_int1 <- df_int1 %>%
  mutate(
    fall_data = as.numeric(as.character(fall_data))
  )


all_pretreat_cov <- c(
  "language_skills",
  "female",
  "ba_quality",
  "extracurricular_skills",
  "ma",
  "same_exp",
```

```
  "exp_highquality"
)


all_pretreat_cov_fall <- c(
  "language_skills",
  "female",
  "ba_quality",
  "extracurricular_skills",
  "ma",
  "same_exp",
  "exp_highquality",
  "fall_data"
)



forest2df_X <- df_int1 %>%
  select(all_of(all_pretreat_cov_fall))%>%
  as.matrix()

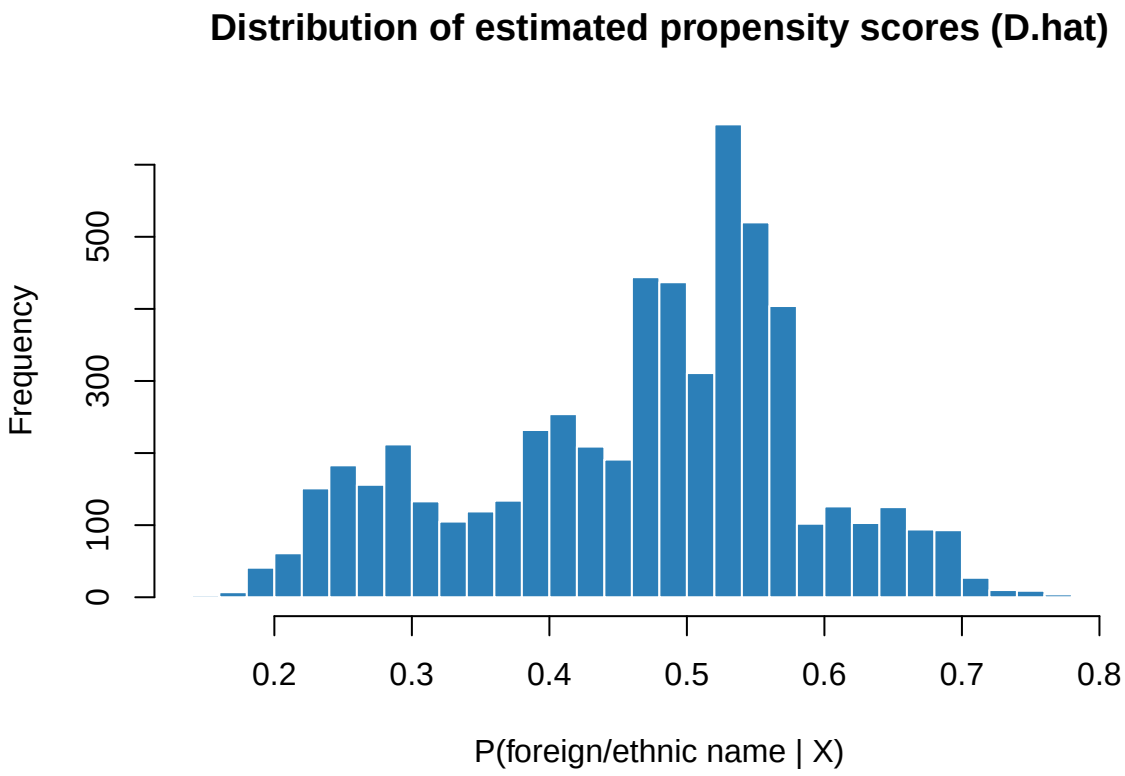
forest2df_treatment <- df_int1$D %>% as.matrix()
forest2df_cluster<-df_int1$firmid %>% as.matrix()
forest2df_Y<-df_int1$callback %>% as.matrix()

cf2<-causal_forest(forest2df_X,
                   forest2df_Y,
                   forest2df_treatment,
                   clusters=forest2df_cluster)
```

Overlap / propensity check

Because name ethnicity is experimentally randomized within a job posting, the estimated propensity scores should cluster fairly tightly around the actual treatment share rather than displaying the extreme values that signal confounding.

```
hist(cf2$W.hat,
     breaks = 30,
     col = "#2c7fb8",
     border = "white",
     main = "Distribution of estimated propensity scores (D.hat)",
     xlab = "P(foreign/ethnic name | X)")
```



Average treatment effect

```
ate <- average_treatment_effect(cf2, target.sample = "all")
ate
```

```
##     estimate      std.err
## -0.044552640  0.007433391
```

```
cat(sprintf(
  "**Estimated ATE:** having a foreign/ethnic-sounding name changes the callback probability by **%.3f*
  100 * ate["estimate"], 100 * ate["std.err"]
))
```

Estimated ATE: having a foreign/ethnic-sounding name changes the callback probability by **-4.455** percentage points (SE = 0.743), on average across the analysis sample.

Variable importance

Which of the three moderators does the forest use most to split on (i.e., which one drives the most predictable heterogeneity)?

```r
vi <- variable_importance(cf2)
vi_tab <- data.frame(
  variable = colnames(forest2df_X),
  importance = as.numeric(vi)
) %>% arrange(desc(importance))

kable(vi_tab, digits = 3, caption = "Causal forest variable importance")
```

Table 2: Causal forest variable importance

variable	importance
ba_quality	0.202
female	0.201
extracurricular_skills	0.135
language_skills	0.107
exp_highquality	0.092
same_exp	0.088
ma	0.088
fall_data	0.088